2025 International Workshop

Data Science & AI

for the Sustainable Food-Energy-Water Nexus



October 13-14, 2025

Cinvestav, Mexico City

Ciencia y Tecnología
Secretaría de Ciencia, Humanidades, Tecnología e Innovación

Cinvestav

Sociedad Mexicana de Inteligencia Artificial

Welcome to 2025 International Workshop on Data Science and AI for the Sustainable Food-Energy-Water Nexus

Welcome to the 2025 International Workshop on Data Science and AI for the Sustainable Food-Energy-Water Nexus (FEW-AI 2025)!

We are honored to gather such a distinguished assembly of international and national experts to address the most critical sustainability challenges of our time. The interconnected fate of our Food, Energy, and Water (FEW) systems demands immediate, intelligent action, and this workshop is designed to chart that path.

Over the next two days, we will delve into powerful, applied data science and AI solutions. Our plenary talks will showcase direct, high-impact applications. Dr. Yu Ding will illuminate how Data Science for Wind Energy is optimizing performance and reliability, while Dr. Kai Sonder will provide essential insights into AI applications in agricultural research addressing disease risk and climate change impact on crops. Furthermore, Dr. Xiaoou Li will tackle the critical challenge of Navigating the Data Gaps in time series forecasting, a core issue for resilient energy management. Dr. Juan Humberto Sossa Azuela's discussion on AI, the Metaverse, and Education, will tackle the ethical, curriculum, and governance imperatives necessary to cultivate the next generation of *AI-literate* scientists and engineers

Crucially, we recognize that technological advancement requires a prepared society and a resilient economy capable of absorbing and scaling these complex innovations. Our robust panel discussions are intentionally designed to broaden the scope beyond pure technical research, focusing instead on the holistic ecosystem needed for successful national AI adoption. The panel on "Opportunities of AI Application in Mexico" will offer a pragmatic view of this transition, detailing how top researchers are already implementing solutions. This segment will specifically explore how industrial automation, advanced health robotics, and sophisticated data mining techniques are driving measurable local economic and social growth within critical sectors like manufacturing and public welfare. This leads directly into our second major discussion, the "Education in the AI Era" panel, which addresses the necessary ethical and curricular shifts. This vital discussion will tackle the governance challenges, ethical frameworks, and necessary curriculum changes to ensure that academic institutions are actively cultivating the next generation of AI-literate talent required to sustain this technological leap.

This workshop is a true nexus point for ideas, connecting advanced research with practical, real-world FEW applications, and innovation with ethical oversight. We encourage you to engage actively, collaborate fiercely, and leave inspired to apply the power of Data Science and AI toward a more sustainable and resilient future for all. Thank you for joining us.

Xiaoou Li
General Chair
Cinvestav
Mexico City, Mexico

Program

October 13, 2025

Time	Activity	Speakers
10:00 - 10:30	Opening Session	
10:30 - 11:30	Keynote 1 ***Data Science for Wind Energy***	Dr. Yu Ding
11:30 - 12:00	Coffee Break	
12:00 - 13:00	Keynote 2 ***AI, Metaverse and Education***	Dr. Juan Humberto Sossa Azuela
13:00 - 15:00	Lunch	
15:00 - 16:30	Panel 1 ***Education in the AI Era***	Dra. Lourdes Martínez Villaseñor Dr. Juan Humberto Sossa Azuela Dra. Xiaoou Li Dr. Genaro J. Martínez Dr. Joselito Medina Marín Dra. Lisbeth Rodríguez-Mazahua

October 14, 2025

Time	Activity	Speakers
10:30 - 11:30	Keynote 3 ***Brave New World, Use of AI in Agricultural Research***	Dr. Sc. Agr. Kai Sonder
11:30 - 12:00	Coffee Break	
12:00 - 13:00	Keynote 4 ***Navigating the Data Gaps: Time Series Forecasting with Incomplete Information***	Dr. Xiaoou Li
13:00 - 15:00	Lunch	
15:00 - 16:30	Panel 2 ***Opportunities for AI Application in Mexico***	Dr. Giner Alor Hernández Dra. Bella Citlali Martinez Seis Dr. Mario César Maya Rodríguez Dr. Hipolito Aguilar Sierra Dr. Andrés José Rodríguez Torres Dr. Luis Arturo Soriano Avendaño

Plenary Talks

October 13, 2025, 10:30 - 11:30

Keynote 1:

Data Science and Wind Energy

Yu Ding

Georgia Institute of Technology, USA

The speaker published a book in 2019, entitled 'Data Science for Wind Energy,' in which a set of wind operational problems is framed through a general formulation of f(Y|X), where Y is the response on a turbine and X is the wind and environmental input. When Y is the wind power output, the conditional density, f(Y|X), is related to the power curve model of wind turbines and can be used to characterize a wind turbine's performance and production efficiency. When Y is the mechanical load, the conditional density, f(Y|X), is related to the load estimation and reliability management for the wind turbines. Modeling and understanding the dynamic in X itself is also important, because a good model of X leads to better wind forecasting. The book is therefore structured in three parts: wind field modeling for forecasting, power curve modeling for performance analysis, and turbine reliability and fault detection. In this talk, the speaker will discuss work related to the second part of the book and present examples in which data science solutions make sensible impact.



Dr. Yu Ding is the Anderson-Interface Chair and Professor in the H. Milton Stewart School of Industrial and Systems Engineering at Georgia Tech. Prior to joining Georgia Tech, he was the Mike and Sugar Barnes Professor of Industrial and Systems Engineering at Texas A&M University. Dr. Ding's research focuses on data and quality science and system informatics. He is the author of the CRC Press book, Data Science for Wind Energy, and a co-author of the Springer Nature book, Data Science for Nano Image Analysis. His research work is recognized by the 2019 IISE's Technical Innovation Award, 2022 INFORMS' Impact Prize, 2024 ASME's Blackall Machine Tool and Gage Award, 2024 SME's S. M. Wu Research Implementation Award, and 2024 IISE Energy Systems Division's Career Achievement Award. Dr. Ding is Senior Vice President of IISE on International Operations and the Editor-in-Chief of *INFORMS Journal on Data Science*. Dr. Ding served as Program Chair for *IEEE CASE* 2025 and the Editor-in-Chief of *IISE Transactions* for the term of 2021-2024.

October 13, 2025, 12:00 - 13:00

Keynote 2:

AI, Metaverse and Education

Juan Humberto Sossa Azuela

Computing Research Center, National Polytechnic Institute (CIC-IPN), Mexico

Artificial Intelligence is the ability of machines to imitate intelligent human behaviors and develop human-like tasks. It combines algorithms that allow computers to: 1) Perceive their environment, 2) Learn from perceived data, 3) Solve problems, and 4) Make appropriate decisions, with the objective of carrying out functions that until now were only achievable by human beings. A Metaverse, on the other hand, is a shared digital world, where, as users, we can travel and interact through our own avatar in an immersive three-dimensional virtual world. Within a metaverse we can carry out a great variety of activities such as socializing, playing, working, shopping, remotely attending concerts or sporting events, etc. In this talk we will address some details about the horizontal technology of Artificial Intelligence, as well as about the metaverse and what role Artificial Intelligence plays in its establishment. Details will be given about the applications that are coming and the necessary requirements for establishment and acceptance of what in the near future will be the next digital leap of our humanity.



Dr. Juan Humberto Sossa Azuela es doctor en Informática por el Instituto Nacional Politécnico de Grenoble, Francia. Es profesor de tiempo completo del Instituto Politécnico Nacional y Director del Centro de Investigación en Computación. Es miembro Emérito del Sistema Nacional de Investigadores. Es miembro de la Academia Mexicana de Ciencias y miembro de la Academia de Ingeniería. Es también miembro Senior del Instituto de Ingenieros Eléctricos y Electrónicos (IEEE) y de la Sociedad Internacional de Redes Neuronales (INNS). Es miembro de la Academia de Maquinaria Computacional (ACM) y Fellow de la Sociedad Mexicana de Inteligencia Artificial (SMIA). En 2021 fue galardonado con el Premio Nacional de Computación por parte de la Academia Mexicana de la Computación (AMEXCOMP). En 2023 fue galardonado con el Premio a la Investigación en Instituto Politécnico Nacional en el area de investigación básica. En 2024 recibió por parte del Tecnológico de Estudios Superiores de Ecatepec un Doctorado Honoris Causa. Es autor de 5 libros de texto, 13 patentes, 32 derechos de autor y más de 500 trabajos de congreso y revista. Ha impartido más de 570 pláticas por invitación. Sus áreas de investigación son en Inteligencia Artificial, Aprendizaje para Máquinas, Redes Neuronales Artificiales, Análisis de Imágenes, Reconocimiento de Patrones, Robótica y Metaversos.

October 14, 2025, 10:30 - 11:30

Keynote 3:

Brave new world, use of AI in agricultural research

Kai Sonder

International Maize and Wheat Improvement Center (CIMMYT), Mexico

Artificial Intelligence (AI) is rapidly transforming agriculture, from research to farm support. Within the CGIAR network, machine learning is applied to extrapolate soil, climate, and pest data with greater accuracy, improving soil maps, fertilizer recommendations, and climate adaptation strategies. Combined with advances in remote sensing and satellite imagery, AI enables near real-time insights on crop types, land-use change, and early detection of plant diseases, with hyperspectral data offering new opportunities to monitor traits and variety adoption. At CIMMYT, AI is also accelerating the analysis of the world's largest maize and wheat gene bank, helping identify traits like drought tolerance and disease resistance, and reducing the time and cost of breeding new resilient varieties. In parallel, large language models (LLMs) are being explored as cost-effective tools for agricultural extension, delivering site-specific advice to farmers, including those in remote areas. Yet, as adoption spreads, ethical challenges remain concerning energy use, environmental footprint, and data quality, which must be carefully addressed to ensure sustainable and responsible AI deployment.



Dr. Sc. Agr. Kai Sonder holds a PhD in Agricultural Sciences from the University of Hohenheim, Stuttgart, Germany, based on research conducted in soil conservation and erosion modeling on the southern Andean slopes of Colombia during a stay at the International Center for Tropical Agriculture (CIAT) in Cali, Colombia. He also received a Master's degree in tropical plant breeding and agroecology from Justus-Liebig University in Giessen, Germany. Kai Sonder leads CIMMYT's Geographic Information Systems (GIS) Unit as a Spatial Data Scientist. His work focuses on technology targeting, crop disease risk modeling, and the impact of climate change on maize and wheat production systems. He was previously the manager of the Geospatial Laboratory at the International Institute of Tropical Agriculture (IITA) in Ibadan, Nigeria. This work focused primarily on spatial analysis of maize, cassava, cowpea, yam, banana/plantain, and soybean crops in Sub-Saharan Africa, as well as on the socioeconomics of the value chains and impacts of these crops. She subsequently completed a post-doctoral fellowship at the International Livestock Research Institute (ILRI) based in Addis Ababa, Ethiopia. Her areas of work included fodder trees in semi-arid and highland valley areas, and geospatial analysis of animal productivity and water use in the Nile Basin. She also worked on GIS applications for the Forage Genetic Resources Unit and Animal Genetic Resources Projects at ILRI.

October 14, 2025, 12:00 - 13:00

Keynote 4:

Navigating the Data Gaps: Time Series Forecasting with Incomplete Information

Xiaoou Li

Computer Science Department, CINVESTAV-IPN, Mexico

Time series forecasting is a cornerstone of modern data-driven decision-making, from predicting stock prices to managing energy consumption. However, the integrity of these forecasts is critically challenged by a pervasive and often overlooked issue: missing data. This is particularly pronounced in real-world systems like smart buildings, where sensor failures, network outages, and other disruptions lead to incomplete time series. This talk will explore the fundamental problem of building resilient forecasting models that can handle these data gaps. While drawing from our recent work on robust electricity forecasting using a Concept Echo State Network (CESN), we will generalize the discussion to a broader framework applicable across various domains. We will present a novel approach that not only maintains high accuracy but also delivers meaningful insights, even when faced with imperfect information. This research aims to provide a path forward for advancing data-driven decision-making in a world where data is often incomplete.



Dr. Xiaoou Li is a Researcher and Professor in the Computer Science Department at CINVESTAV-IPN in Mexico. She has also been a Senior Research Fellow at Queen's University Belfast, the University of California Santa Cruz, Shenzhen University, and Northeastern University for her sabbatical leaves. Dr. Li is a member of the Mexican Academy of Science and a Level 3 member of the National Researcher System of Mexico (SNII). She is the author/co-author of over 300 publications in international journals, books, and conference proceedings. Her research interests include machine learning, data mining applications, data-driven sustainable and green automation, robot-assisted rehabilitation systems, and learning in technology-rich environments. Dr. Li actively organizes academic events and has served as the General Chair of IEEE conferences such as ICNSC 2016 and CASE 2022, and as the Program Chair of SSCI 2023. She has been a member of various editorial boards, including IEEE Press, Engineering Applications of Artificial Intelligence (EAAI), IEEE Transactions on Automation Science and Engineering (T-ASE), IEEE/CAA Journal of Automatica Sinica (JAS), IEEE Access, and IEEE SMC Magazine. She was a member of IEEE James H. Mulligan, Jr. Education Medal Committee (2019-2023).

Panel Discussions

October 13, 2025, 15:00 - 16:30

Panel 1:

Education in the AI Era

Join six of Mexico's leading researchers in Artificial Intelligence and Computer Science for an in-depth and critical exploration of Education in the AI Era. This high-level panel will move beyond generalities to address the technical, ethical, and practical challenges of integrating AI into the academic landscape.

1. *Ethics, Equity, and Governance of AI in Education*

This segment will delve into the critical ethical and regulatory dimensions of AI integration. Dra. Lourdes Martínez Villaseñor, a recognized expert and award-winner in Responsible AI Ethics (RAI), will lead a discussion on: AI Bias and Equity; Privacy and Transparency; The Educator's Responsibility

2. *Redefining Curriculum, Learning, and Assessment.*

With the rise of Generative AI, traditional methods of teaching and assessment are fundamentally disrupted. The panel will explore how curricula must evolve to maintain rigor in a world of instant answers. Dr. Juan Humberto Sossa Azuela and Dr. Xiaoou Li, experts in Machine Learning, Pattern Recognition, and Learning in Technology-Rich Environments, will discuss which skills- such as creativity, critical thinking, and complex problem-solving- must now be prioritized over rote memorization. The shift from grading final, potentially AI-generated products (like essays) to evaluating the process of learning, including project-based learning and the application of complex algorithms, a topic Dr. Joselito Medina Marín's expertise in Optimization of Systems and Autómata Theory is crucial for.

3. *Technical Integration: Databases, Optimization, and Emerging AI Paradigms*

This section will tackle the technical foundation required to support a modern, AI-enabled educational system, drawing on the panelists' deep research backgrounds. Dr. Lisbeth Rodríguez-Mazahua, a specialist in Autonomic Database Systems and Big Data, will address the challenges of managing, storing, and utilizing vast amounts of student data effectively and securely for personalized learning systems. Dr. Genaro J. Martínez, known for his work in Cellular Automata and Complex Systems. Dr. Sossa Azuela and Dr. Li will discuss the technical feasibility and impact of deploying Intelligent Tutoring Systems (ITS) and adaptive learning platforms that tailor content to the individual student's pace and needs using cutting-edge AI.

Panelists:

Dra. Lourdes Martínez Villaseñor
Dra. Xiaoou Li
Dr. Joselito Medina Marín
Dr. Juan Humberto Sossa Azuela
Dr. Genaro J. Martínez
Dra. Lisbeth Rodríguez-Mazahua



Dra. Lourdes Martínez Villaseñor es Ingeniera en Sistemas Computacionales por el Tecnológico de Monterrey Campus Estado de México, Maestra en Educación Familiar por el IPCE de la Universidad Panamericana y Doctora en Ciencias Computacionales del Tecnológico de Monterrey, Campus Estado de México. Es profesora investigadora nivel D en el área de inteligencia artificial en la Universidad Panamericana y jefa académica de la Maestría en Ciencia de Datos. Tiene la distinción de nivel 1 del Sistema Nacional de Investigadores del Conahcyt y de Profesor investigador nivel D de la Universidad Panamericana. Es Presidenta de la Sociedad Mexicana de Inteligencia Artificial. Ganó el premio de Mujer Líder en Inteligencia Artificial Responsable en Norteamérica 2023. Este año es ganadora también del Trailblazer Award – Woman in AI of the Year de Norteamérica 2025. Sus principales intereses de investigación son la inteligencia artificial aplicada al cuidado de la salud, el modelado de usuarios, y la ética de la inteligencia artificial. Ha sido docente por más de 32 años y autora de más de 65 publicaciones en revistas y conferencias internacionales, capítulos de libros y libros en el área de inteligencia artificial.



Dr. Juan Humberto Sossa Azuela es doctor en Informática por el Instituto Nacional Politécnico de Grenoble, Francia. Es profesor de tiempo completo del Instituto Politécnico Nacional y Director del Centro de Investigación en Computación. Es miembro Emérito del Sistema Nacional de Investigadores. Es miembro de la Academia Mexicana de Ciencias y miembro de la Academia de Ingeniería. Es también miembro Senior del Instituto de Ingenieros Eléctricos y Electrónicos (IEEE) y de la Sociedad Internacional de Redes Neuronales (INNS). Es miembro de la Academia de Maquinaria Computacional (ACM) y Fellow de la Sociedad Mexicana de Inteligencia Artificial (SMIA). En 2021 fue galardonado con el Premio Nacional de Computación por parte de la Academia Mexicana de la Computación (AMEXCOMP). En 2023 fue galardonado con el Premio a la Investigación en Instituto Politécnico Nacional en el area de investigación básica. En 2024 recibió por parte del Tecnológico de Estudios Superiores de Ecatepec un Doctorado Honoris Causa. Es autor de 5 libros de texto, 13 patentes, 32 derechos de autor y más de 500 trabajos de congreso y revista. Ha impartido más de 570 pláticas por invitación. Sus áreas de investigación son en Inteligencia Artificial, Aprendizaje para Máquinas, Redes Neuronales Artificiales, Análisis de Imágenes, Reconocimiento de Patrones, Robótica y Metaversos.



Dr. Xiaoou Li is a Researcher and Professor in the Computer Science Department at CINVESTAV-IPN in Mexico. She has also been a Senior Research Fellow at Queen's University Belfast, the University of California Santa Cruz, Shenzhen University, and Northeastern University for her sabbatical leaves. Dr. Li is a member of the Mexican Academy of Science and a Level 3 member of the National Researcher System of Mexico (SNII). She is the author/co-author of over 300 publications in international journals, books, and conference proceedings. Her research interests include machine learning, data mining applications, data-driven sustainable and green automation, robot-assisted rehabilitation systems, and learning in technology-rich environments. Dr. Li actively organizes academic events and has served as the General Chair of IEEE conferences such as ICNSC 2016 and CASE 2022, and as the Program Chair of SSCI 2023. She has been a member of various editorial boards, including IEEE Press, Engineering Applications of Artificial Intelligence (EAAI), IEEE Transactions on Automation Science and Engineering (T-ASE), IEEE/CAA Journal of Automatica Sinica (JAS), IEEE Access, and IEEE SMC Magazine. She was a member of IEEE James H. Mulligan, Jr. Education Medal Committee (2019-2023).



Dr. Genaro J. Martínez is a full-time researcher at the National Polytechnic Institute in Mexico City. He is an active member of several institutions and laboratories around the world, such as the International Unconventional Computing Laboratory at the University of the West of England in the United Kingdom; the Foundation of Computer Science Laboratory at the Hiroshima University in Japan; the Center for Chaos and Complex Networks at the City University of Hong Kong in China; the Unconventional Algorithms and Computing Lab at the Technical University of Ostrava in Czech Republic; the Laboratoire de Recherche Scientifique in France, and the Institute des Systemes Complexes en Normandie in France. Prof. Martínez is an editorial member of the Journal of Cellular Automata, Complex Systems, Physica D, International Journal of Bifurcation and Chaos, Artificial Life, Nonlinear Dynamics, Chaos, Information Sciences, International Journal of Natural Computing Research, IEEE Transactions on Circuits and Systems, and the Mathematical Reviews, American Mathematical Society. Prof. Martínez together with prof. Adamatzky had published the book “Designing Beauty: The Art of Cellular Automata”. Also, he is leading two computational paradigms in cellular automata theory: colliders cellular automata and computation by competing patterns with cellular automata.



Dr. Joselito Medina Marín recibió el grado de Ingeniero en Computación por parte de la Universidad Autónoma de Guerrero en el año de 1997. En el 2002 recibió el grado de Maestro en Ciencias con especialidad en Ingeniería Eléctrica opción Computación, y en el 2005 el grado de Doctor en Ciencias con especialidad en Ingeniería Eléctrica opción Computación, ambos grados fueron otorgados por el Centro de Investigación y de Estudios Avanzados del Instituto Politécnico Nacional (CINVESTAV – IPN). Ha publicado más de 20 artículos en revistas indexadas por el Journal Citation Reports. Ha presentado más de 20 ponencias en congresos internacionales. Además, cuenta con 5 capítulos de libro en publicaciones de Springer. Es coautor de los libros "Solución de Problemas en Ingeniería con Matlab", "Investigación de Operaciones", y del libro "Introducción a Matlab para Resolver Problemas de Ingeniería Aplicando Algoritmos Genéticos". Ha dirigido más de 10 tesis de la Licenciatura en Ingeniería Industrial, Maestría de Ciencias en Ingeniería Industrial, Maestría en Ciencias en Ingeniería de Manufactura, y en el Doctorado en Ciencias en Ingeniería Industrial. Forma parte del Sistema Nacional de Investigadores, con nivel 1. Las líneas de investigación que cultiva incluyen análisis, modelación y optimización de sistemas de ingeniería, teoría de Autómatas, Redes de Petri, bases de datos activas y simulación de eventos discretos. Y actualmente es el Subidrector de Ediciones y Publicaciones de la UAEH.



Dr. Lisbeth Rodríguez-Mazahua got a Ph.D. in Computer Science at Centro de Investigación y de Estudios Avanzados del Instituto Politécnico Nacional (CINVESTAV-IPN) in 2012. Since 2014, she has been a Professor of Computer Science at Instituto Tecnológico de Orizaba, Veracruz, Mexico. Her current research interests include distribution design of databases, database theory, autonomic database systems, multimedia databases, data mining, and big data. She has more than 110 publications in scientific journals, books, and conference proceedings about these topics. She has also led six projects funded by the National Council of Humanities, Sciences, and Technologies (CONAHCYT), the Public Education Secretariat (SEP), and the National Technology of Mexico. She has supervised 13 postgraduate students. Since 2016, she has been a member of the National Researcher System of Mexico.

October 14, 2025, 15:00 - 16:30

Panel 2:

Opportunities for AI Application in Mexico

Discover the tangible future of Artificial Intelligence in Mexico with a panel of six distinguished researchers who are actively translating cutting-edge AI concepts into real-world applications across industry, health, and agriculture.

1. *AI in Industrial Automation and Advanced Control*

Drawing on vast expertise in control systems and industrial environments, the panel will detail how AI and Machine Learning are revolutionizing Mexican industry. Dr. Mario César Maya Rodríguez, a specialist in Artificial Neural Networks and Transfer Learning for industrial processes, will discuss using AI to optimize production, predict equipment failure, and implement advanced control systems. Contributions from Dr. Hipolito Aguilar Sierra and Dr. Andrés José Rodríguez Torres, with their extensive backgrounds in Mechatronics, Automatic Control, and Industrial Automation, will highlight the development and implementation of intelligent robotics and automation solutions for greater efficiency and precision.

2. *Social Impact and AI for Public Good*

The discussion will spotlight AI applications focused on improving quality of life and addressing societal needs in Mexico, moving beyond abstract theory to practical solutions. Dra. Bella Citlali Martinez Seis, a leading researcher in Data Mining and Social Network Analysis, will present projects focused on technological developments for health systems and communities with disabilities, emphasizing human-centered design. Dr. Aguilar Sierra will share insights from his work in Robotics & AI Solutions iLab on medical robotics, including the development of exo-skeletons and rehabilitation systems. Additionally, Dr. Luis Arturo Soriano Avendaño, with expertise in Renewable Energy and Agricultural Robotics, will explore how AI-driven modeling and control systems are optimizing resource use in the crucial agricultural sector.

3. *Applied Research, Innovation, and Ecosystem Development*

This segment will focus on the development pipeline from research publication to technological commercialization. Dr. Giner Alor Hernández, a prolific author and editor in the field of Computer Science and AI, will discuss the role of high-impact scientific publication in fostering technological transfer and strengthening national research capacity. The panelists, with their established records of industry experience, CONAHCyT projects, and international research stays (France, Spain, USA), will discuss strategies for strengthening academic-industry linkages and ensuring that Mexican AI talent is prepared to lead innovation at a global level.

Panelists:

Dr. Giner Alor Hernández
Dr. Mario César Maya Rodríguez
Dr. Andrés José Rodríguez Torres
Dra. Bella Citlali Martinez Seis
Dr. Hipólito Aguilar Sierra
Dr. Luis Arturo Soriano Avendaño



Dr. Giner Alor Hernández es Licenciado en Ciencias de la Computación por la Benemérita Universidad Autónoma de Puebla, es Maestro y Doctor en Ciencias por el Centro de Investigación y de Estudios Avanzados del I.P.N. Departamento de Ingeniería Eléctrica, Sección Computación. Es Profesor Investigador de la División de Estudios de Posgrado e Investigación del Instituto Tecnológico de Orizaba, es Miembro del Sistema Nacional de Investigadores nivel 3 del área VIII Ingenierías ante CONACYT y Profesor con reconocimiento de Perfil Deseable ante PRODEP. Cuenta con más de 300 publicaciones en revistas con arbitraje estricto, congresos nacionales e internacionales. Es autor/editor de 8 libros en el área de las ciencias computacionales. Es autor de más de 40 capítulos de libro publicados en editoriales de prestigio. Ha sido editor invitado en más de 14 revistas indizadas en el Journal Citation Reports. Es editor asociado de las revistas IEEE Latin America Transactions y Frontiers in Artificial Intelligence, ambas indizadas en el Journal Citation Reports.



Dra. Bella Citlali Martinez Seis es Profesor Titular de UPIITA-IPN (Instituto Politécnico Nacional). Recibió el doctorado en Ciencias de la Computación del CINVESTAV-IPN, obtuvo la Maestría en el CIC-IPN y es egresada de ESCOM-IPN. Pertenece al Sistema Nacional de Investigadoras e Investigadores (SNII) y es parte de la mesa directiva de la Sociedad Mexicana de Inteligencia Artificial (SMIA). Sus intereses incluyen Minería de Datos, Ciencia de Datos, Inteligencia Artificial y el Análisis de Redes Sociales. Tiene publicaciones en congresos y revistas internacionales, ha dirigido proyectos de investigación en el IPN, ha colaborado en proyectos CONAHCyT con universidades nacionales y extranjeras, ha asesorado a tesistas de licenciatura y ha impartido y cursado diplomados relacionados con competencias docentes. Tiene vinculación y desarrollos tecnológicos enfocado a comunidades con discapacidad auditiva, salud y sistemas socioecológicos.



Dr. Mario César Maya Rodríguez obtuvo la Licenciatura en Ingeniería en Control y Automatización por el Instituto Politécnico Nacional (IPN), Ciudad de México, en 2015. Posteriormente, cursó la Maestría y el Doctorado en Control Automático en el Centro de Investigación y de Estudios Avanzados del IPN (CINVESTAV-IPN), concluidos en 2017 y 2022, respectivamente. Sus líneas de investigación incluyen redes neuronales artificiales, aprendizaje por transferencia, meta-aprendizaje, meta-transferencia de aprendizaje, así como identificación y control avanzado de procesos industriales. Cuenta con amplia experiencia en entornos industriales, abarcando el diseño, implementación y documentación de sistemas de control, incluyendo: Programación de dispositivos de automatización; Selección de equipos de control e instrumentación; Integración de protocolos de comunicación industrial; Diseño, construcción y puesta en marcha de estaciones de control. Desde 2023, es reconocido como Investigador Nacional Nivel 1 por el Sistema Nacional de Investigadores (SNI).



Dr. Hipolito Aguilar Sierra es Ingeniero en Mecatrónica por la UPIITA del IPN. Realizó sus estudios de maestría y doctorado en el Departamento de Control Automático del CINVESTAV-IPN, Unidad Zacatenco. Su trayectoria académica incluye cargos como profesor de asignatura en la Universidad del Valle de México Campus San Rafael (2015–2016) y Profesor Investigador Titular B de Tiempo Completo en la Universidad Politécnica de Tulancingo (UPT), donde también fue Jefe del Cuerpo Académico en Instrumentación y Control (2017–2018). Desde 2020, se desempeña como Maestro de Tiempo Completo Investigador B en la Universidad La Salle México, adscrito a la Vicerrectoría de Investigación, donde actualmente lidera el Robotics & AI Solutions iLab, laboratorio enfocado en el desarrollo de soluciones robóticas para la salud, la inclusión y la automatización inteligente. Desde 2025, asumió también la Jefatura del Departamento de Ingeniería, impulsando estrategias de innovación, vinculación y formación en los programas de ingeniería de la institución. Es miembro del Sistema Nacional de Investigadores con el nivel I (2021–2025), ha dirigido tres tesis de maestría y una de doctorado, y cuenta con más de 200 citas académicas. Ha realizado estancias de investigación en el laboratorio Heudiasyc de la Universidad Tecnológica de Compiègne (Francia), en el Centro de Investigación en Ingeniería Mecánica de la Universidad Politécnica de Valencia (España) y en el Laboratorio de Robótica Existencial de la Universidad de California en San Diego (EE. UU.). Sus líneas de investigación se centran en el control automático de robots manipuladores paralelos, el diseño de observadores para diagnóstico de fallas en sistemas no lineales, así como en el desarrollo de exoesqueletos y sistemas de robótica médica orientados a la rehabilitación y el aumento de capacidades físicas.



Dr. Andrés José Rodríguez Torres is a professor and researcher of ITAM - Instituto Tecnológico Autónomo de México, in the fields of automatic control, industrial automation, and mechatronics systems, with extensive experience in developing technological innovation projects, conducting research both nationally and internationally, and providing business consulting services. I am dedicated to educating undergraduate and graduate students, focusing on practical application and scientific advancement to positively impact the academic community and technological development. I have a significant publication record in the areas of vibration control, underactuated systems, and renewable energy, and my multilingual skills facilitate international collaboration.



Dr. Luis Arturo Soriano Avendaño es profesor investigador del Departamento de Ingeniería Mecánica Agrícola, Universidad Autónoma Chapingo. Doctorado en Ciencias con Especialidad en Control Automático. SNII Nivel I. Su preparación académica incluye: Ingeniería en Robótica Industrial, una Maestría en Ingeniería de la Manufactura con especialidad en Control Automático ambas por la Escuela Superior de Ingeniería Mecánica y Eléctrica del Instituto Politécnico Nacional Unidad Azcapotzalco, Recibió el grado de Doctor en Ciencias con especialidad en Control Automático por el Centro de Investigación y de Estudios Avanzados del Instituto Politécnico Nacional. Eres Posdoctorante de la Escuela de Ingeniería y Ciencias del Tecnológico de Monterrey. Además es miembro del Sistema Nacional de Investigadores de México (Nivel Candidato). Cuenta con más de 9 años de experiencia en la industria. Se desempeñó como Administrador y Desarrollador de proyectos de Tecnologías de la Información y Sistemas embebidos en la empresa ADO y Empresas Coordinadas. Desarrollador de aplicaciones a la medida para Intelisis, también fue profesor del Departamento de Electrica Electronica de ESIME Azcapotzalco del IPN. Cuenta con artículos en revistas internacionales y conferencias internacionales, en el área de modelado matemático y control para Energías Renovables y Robótica.

FEW-AI Lab

Machine Learning and Big Data Analytics for Sustainable Food-Energy-Water (FEW) Nexus





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